Yulei Fu

US Citizen | yulei-machine-shop.com | yuleifu@umich.edu | linkedin.com/in/yulei-fu | 7342394771

EDUCATION

University of Michigan Sep 2023 - May 2027

Bachelors of Science in Robotics Engineering (Mechatronics) Ann Arbor, MI

Bachelors of Science in Computer Science

SKILLS

Languages: C, C++, Python, Bash

Software Tools: ZephyrOS, FreeRTOS, gdb, git, Makefile, Docker, Google Test, CMake, Yocto, Jira

Relevant Coursework: Robot Operating Systems (ROS2), Logic Design, Data Structures & Algorithms, Embedded Systems

EXPERIENCE

Amazon Robotics Jan 2025 - June 2025

Embedded Software Engineer Co-Op Westborough, MA

- Designed and owned a C++ logging and system log capture pipeline for a safety-certified controller to replace an ad-hoc debugging process, cutting issue-reproduction time by 85% (6 hrs to 0.75 hrs).
- Developed a robust multithreaded D-Bus service for embedded Linux safety systems, aggregating logs from multiple subsystems with safe batching, deterministic behavior, and low-latency IPC.
- Deployed and designed a remote Wi-Fi diagnostics API, adding to the system's hardware-diagnostics suite, reducing field-troubleshooting from 2+ hours to 0.03 hours for on-sites that previously had no remote pathway to inspect connectivity issues.
- Architected, implemented, and shipped a telemetry pipeline delivering 10K+ messages/day to AWS S3 over MQTT, cutting data-loss by 94% and enabling a 7-engineer diagnostics team to finally access consistent, fleet-reliable field data.

Stirling Research Group Nov 2024 - Jan 2025

Research Intern Ann Arbor, MI

- Implemented the Keytel metabolic-rate model in JavaScript to convert real-time heart-rate data into energy-expenditure estimates, enabling accurate British Thermal Unit (BTU) tracking for 4+ hour extravehicular activity (EVA) simulations.
- Architected data parsing from wearable heart-rate data at 20-second intervals, applied time-weighted averaging, and updated metabolic-rate estimates with 100% uptime across all test sessions.
- Created the experimenter dashboard with real-time HR monitoring, configurable alert thresholds, labeled task segments, and automatic data export, improving post-session analysis efficiency by 40%.
- Connected the nRF52840 based heart-rate wearable to the testbench via bluetooth.

PROJECT EXPERIENCE

Custom ROS2 Hardware Interface Driver (STM32, UART, ROS2)

- Implemented a custom ROS2 hardware interface to stream IMU data from an STM32F411 microcontroller, defining a binary UART protocol with explicit framing, sequence counters, and embedded microcontroller timestamps.
- Wrote bare-metal C firmware using interrupt-driven UART with DMA buffering to sample and transmit IMU data at 500 Hz, maintaining deterministic timing with <1% jitter measured via hardware timers.
- Measured end-to-end latency by embedding microcontroller timestamps in each frame and comparing against ROS publish time, validating <3 ms mean latency over sustained 10-minute runs.

Real-Time IMU Telemetry + Safety Monitor on MCU (FreeRTOS)

- Developed an end-to-end embedded robotics firmware feature on an ESP32, implementing fixed-rate IMU sensor ingestion, filtering, and real-time telemetry streaming over USB serial to a ground station.
- Enforced deterministic execution using FreeRTOS task scheduling, measuring loop jitter and worst-case execution time while operating under injected system load.